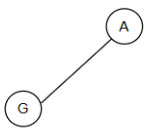
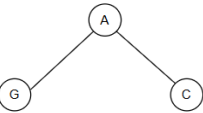
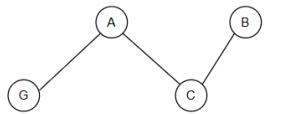
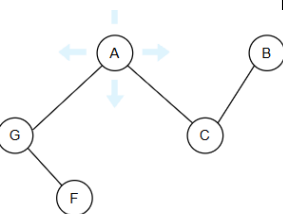
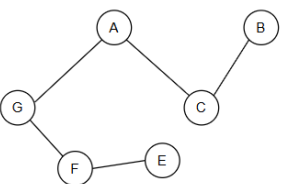
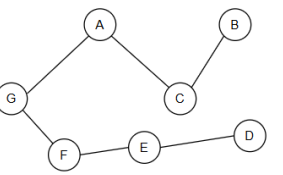
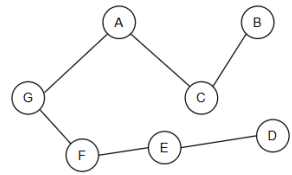


Report Graph Theory

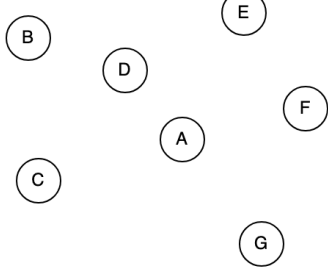
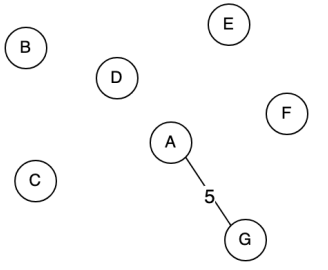
Github Repository link : <https://github.com/rbindech/mst-algorithms-group-9.git>

Prim's algorithm

Step	U	Edge(u.v)	U.V	Graf
0	{ }	-	{A,B,C,D,E,F,G}	-
1	{A}	A-G : 5 A-C : 6 A-B : 7 A-F : 10	{B,C,D,E,F,G}	
2	{A,G}	A-C : 6 A-B : 7 G-F : 6 A-F : 10	{B,D,E,F,G}	
3	{A,G,C}	A-B : 7 G-F : 6 C-F : 9 C-B : 5 C-E : 7 A-F : 10	{B,D,E,F}	
4	{A,G,C,B}	A-B: 7 (cycle), G-F: 6 B-D: 7 C-E: 7 C-F: 9 B-E: 9 A-F: 10	{D, E, F}	
5	{A, G, C, B, F}	F-E: 5 A-B: 7 (cycle) B-D: 7 C-E: 7 C-F: 9 (cycle) B-E: 9 A-F: 10 (cycle)	{D, E}	
6	{A, G, C, B, F, E}	E-D: 5(chosen) A-B: 7 (cycle) B-D: 7 C-E: 7 (cycle) B-E: 9 (cycle) C-F: 9 (cycle)	{D}	

		A-F: 10 (cycle)		
7	{A, G, C, B, F, E, D}	All nodes visited	{ }	

Kruskal's algorithm

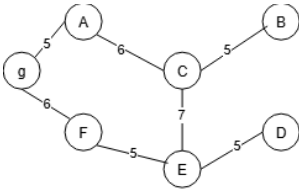
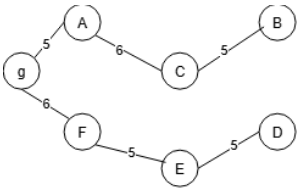
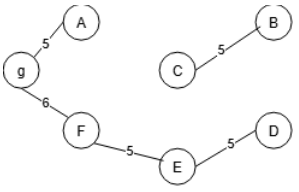
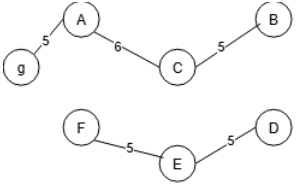
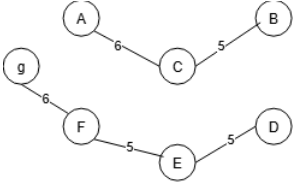
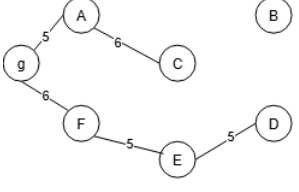
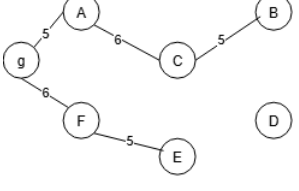
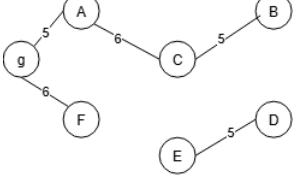
Step	<i>Edges Remaining Sorted</i>	Is the selected edge creating a cycle ?	<i>Connected Components</i>	Graf
0	A-G : 5 B-C : 5 D-E : 5 E-F : 5 A-C : 6 F-G : 6 A-B : 7 B-D : 7 C-F : 7 B-E : 9 C-E : 9 A-F : 10	-	{A}, {B}, {C}, {D}, {E}, {F}, {G}	
1	A-G : 5 V B-C : 5 D-E : 5 E-F : 5 A-C : 6 F-G : 6 A-B : 7 B-D : 7 C-F : 7 B-E : 9 C-E : 9 A-F : 10	Ok, no cycle	{A,G}, {B}, {C}, {D}, {E}, {F}	

2	B-C : 5 V D-E : 5 E-F : 5 A-C : 6 F-G : 6 A-B : 7 B-D : 7 C-F : 7 B-E : 9 C-E : 9 A-F : 10	Ok, no cycle	{A,G}, {B,C}, {D}, {E}, {F}	
3	D-E : 5 V E-F : 5 A-C : 6 F-G : 6 A-B : 7 B-D : 7 C-F : 7 B-E : 9 C-E : 9 A-F : 10	Ok, no cycle	{A,G}, {B,C}, {D,E}, {F}	
4	E-F : 5 V A-C : 6 F-G : 6 A-B : 7 B-D : 7 C-F : 7 B-E : 9 C-E : 9 A-F : 10	Ok, no cycle	{A,G}, {B,C}, {D,E,F}	
5	A-C : 6 V F-G : 6 A-B : 7 B-D : 7 C-F : 7 B-E : 9 C-E : 9 A-F : 10	Ok, no cycle	{A,G,B,C}, {D,E,F}	

6	F-G : 6 V A-B : 7 B-D : 7 C-F : 7 B-E : 9 C-E : 9 A-F : 10	Ok, no cycle	{A,G,B,C,D,E,F} <i>All nodes connected</i>	
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Reverse Delete Algorithm

Step	Edge(v,u) to delete	Are U and V still connected ?	Graf
0	-	-	
1	{A-F} = 10	yes AF still connected	
2	{B-E} = 9	Yes BE still connected	
3	{C-F} = 9	yes CF still connected	
4	{A-B} = 7	Yes AB still connected	

5	$\{B-D\} = 7$	Yes BD still connected	
6	$\{C-F\} = 7$	Yes CF still connected	
7	$\{A-C\} = 6$	No we can't delete AC	
8	$\{F-G\} = 6$	No we can't delete FG	
9	$\{A-G\} = 5$	No we can't delete AG	
10	$\{B-C\} = 5$	No we can't delete BC	
11	$\{D-E\} = 5$	No we can't delete DE	
12	$\{E-F\} = 5$	No we can't delete EF	

13	-	Final Graph	<pre> graph LR g((g)) --- 5 A((A)) A --- 6 C((C)) C --- 5 B((B)) g --- 6 F((F)) F --- 5 E((E)) E --- 5 D((D)) </pre>
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